RFNet-Edit: Fantasy-Command Scene Editing

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***Abstract*—Text-guided image editing aims to modify visual content according to natural-language instructions while preserving scene consistency. Although recent diffusion-based inpainting models have shown promising results, they often struggle with complex instructions involving multiple objects and strict background preservation. In this paper, we present RFNet-Edit, a structured, LLM-assisted image editing framework that decomposes free-form editing instructions into executable, instance-level edit plans. Our approach integrates large language models for instruction parsing with open-vocabulary object detection, pixel-level segmentation, and diffusion-based inpainting. By explicitly modeling edit actions, object boundaries, and background constraints, RFNet-Edit supports multi-object editing and reduces unintended background alterations. We evaluate our method on 95 images from the EditBench dataset using CLIP score as a measure of semantic alignment. RFNet-Edit achieves a CLIP score of 27.41, outperforming a Stable Diffusion XL inpainting baseline, which attains a score of 25.69 under the same evaluation protocol. Qualitative results further demonstrate improved object coherence and background preservation, highlighting the effectiveness of structured instruction interpretation for controllable image editing. Our code is available at https://github.com/Terence1219/Reality-and-Fantasy-Edit**

***Index Terms*—Text-guided Image Editing, Image Inpainting, Large Language Models (LLMS), Multi-Object Editing, Vision-Language Interaction**

I. Introduction

Text-guided image editing enables users to modify images using natural-language instructions and has become an important problem in vision-language research. Recent diffusion-based models have shown strong performance in image synthesis and inpainting, allowing localized image modifications guided by textual prompts. However, these approaches often struggle when instructions require multiple coordinated edits, precise object-level control, or strict preservation of the original background.

Datasets such as EditBench [5] expose these limitations by introducing complex, realistic editing scenarios that involve multiple atomic actions, instance-level transformations, and explicit constraints on which regions of the image should remain unchanged. Conventional inpainting pipelines typically operate on a single mask and rely on unconstrained text prompts, which can lead to background hallucination, object misplacement, or incomplete execution of the intended edits.

A key challenge lies in the gap between unstructured language instructions and the structured visual operations required for reliable image editing. Without an explicit representation of edit intent—such as target objects, edit actions, and background constraints—vision models must implicitly infer structure, often resulting in unstable or inconsistent outcomes.

Inspired by RFNet [4], leveraging LLMs' reasoning ability in assisting image generation, we introduce RFNet-Edit, a framework that explicitly separates instruction understanding from visual execution. Rather than relying on implicit reasoning within a generative model, RFNet-Edit structures edit intent prior to image synthesis, enabling more reliable object-level control. This design allows the editing process to be decomposed into interpretable stages, facilitating multi-object edits while maintaining background consistency. To implement this, RFNet-Edit automates the editing pipeline through two specialized stages: In-Depth Object Generation and Seamless Background Integration. Unlike conventional inpainting models (e.g., SDXL) that rely on rigid, user-defined masks, our approach automatically infers object localization using bounding boxes to synthesize target objects that are semantically aligned with the edited prompt. To ensure visual coherence, we subsequently employ the Segment Anything Model (SAM) to define precise object boundaries and utilize a latent-freezing technique during the denoising process. By restricting generative modifications to the interfacial regions between the new object and the existing scene, our model achieves a seamless transition while preserving the integrity of the original background.

We validate our approach on the EditBench dataset, demonstrating that structured instruction interpretation leads to improved semantic alignment and visual coherence in challenging editing scenarios involving multiple objects and background constraints.

In summary, this work makes three key contributions: (1) A structured, LLM-based framework that automates the decomposition of complex natural-language instructions into executable, instance-level edit plans, supporting multi-object image editing; (2) A novel two-stage generative architecture—comprising In-Depth Object Generation and Seamless Background Integration—that utilizes bounding boxes for enhanced object flexibility and latent freezing for precise boundary blending; and (3) Comprehensive empirical validation demonstrating that RFNet-Edit achieves superior semantic alignment and background preservation compared to state-of-the-art inpainting baselines.

II. Method

A. System Overview

RFNet-Edit is implemented as a modular, four-stage image editing pipeline designed to execute structured edit plans derived from natural-language instructions. The pipeline consists of the following components:

- Instruction Parsing (LLM)
- Object Localization
- Pixel-Level Segmentation
- Object-Aware Inpainting (RFNet-Edit or SDXL)

The system transforms a natural-language editing instruction into a structured, executable plan that supports instance-level editing and multi-object operations.

Given an input image and a free form of editing instruction, the system first decomposes the instruction into a structured representation describing the intended edits. This representation explicitly specifies target objects, desired transformations, and background preservation constraints. The structured edit plan is then passed to a vision pipeline that localizes and segments the relevant objects, producing precise masks that guide localized image synthesis. Finally, the system performs object-aware inpainting using either a standard diffusion-based inpainting model (SDXL) [3] or the proposed RFNet-Edit pipeline, ensuring that modifications are confined to the intended regions while preserving the original background. This modular design allows each stage to be independently improved or replaced and enables robust handling of complex instructions involving multiple objects and actions.

B. LLM-Based Instruction Parsing

To bridge the gap between unconstrained natural language and executable image editing operations, we employ GPT-4o-mini [6] as an instruction parsing module. The LLM converts user-provided editing instructions into a standardized JSON schema that serves as the interface between language understanding and vision-based processing. Unlike prior approaches that assume a single edit per instruction, our schema supports multiple atomic edits, enabling the system to handle compound commands such as replacing one object while simultaneously modifying another. Each atomic edit contains the following fields:

- target object (the object to be edited)
- output object (the desired object after editing)
- action (replace, modify, remove)
- generation prompt (a concise text prompt used to guide image synthesis)
- background description (a textual constraint describing the surrounding scene)
- preserve subject (a Boolean flag indicating item descriptions (one concise and two expanded variants to improve robustness downstream generation)

This structured representation reduces ambiguity and enforces consistency with EditBench annotations, allowing the vision modules to operate deterministically. In addition, separating high-level intent (action and target) from low-level generation prompts to improve controllability and makes the system more interpretable and debuggable.

C. Object Detection with GroundingDINO

For object localization, we adopt GroundingDINO-SwinT [1], an open-vocabulary object detection model capable of predicting bounding boxes directly from textual queries. Given the target_object field produced by the LLM, GroundingDINO performs text-to-region matching to identify candidate object locations within the image.

This approach avoids reliance on closed-set object detectors and allows RFNet-Edit to generalize to previously unseen object categories. The predicted bounding boxes provide coarse spatial localization, which is sufficient for identifying the region of interest but not precise enough for high-quality inpainting. Therefore, these bounding boxes are used as initialization cues for subsequent pixel-level segmentation rather than as final edit masks.

D. Pixel-Level Segmentation Using SAM

To obtain accurate object boundaries, we refine the detected bounding boxes using the Segment Anything Model (SAM, ViT-H) [2]. For each bounding box, SAM produces a high-resolution binary segmentation mask that captures the precise shape of the target object.

SAM is applied independently to each bounding box, and the resulting masks are combined using logical OR operations to support multi-object editing scenarios. This instance-level segmentation step is crucial for confining image synthesis strictly to the intended object regions and preventing unnecessary modifications to the background.

Because SAM is category-agnostic and does not rely on object-specific training, it integrates seamlessly with the open-vocabulary detection provided by GroundingDINO, forming a flexible and generalizable localization–segmentation pipeline.

E. Output Assembly and Visualization

For each parsed edit instruction, the system assembles all intermediate and final outputs into a structured format suitable for both downstream inpainting and evaluation. Specifically, the following components are produced:

- Bounding boxes
- Segmentation masks
- Generation prompts and descriptions
- Background preservation constraints

All results are stored in a standardized JSON file editbench_llm_results.json, enabling reproducibility and systematic analysis. In addition, the system generates visualizations comparing the original image, EditBench ground-truth masks, and SAM-generated masks. These visual outputs facilitate qualitative inspection, debugging, and error analysis, and are saved alongside the numerical results in the corresponding results/ directory.

This structured output design allows RFNet-Edit to serve not only as an end-to-end editing system but also as a flexible research framework for studying the interaction between language understanding, visual grounding, and generative image editing.

F. Image Inpainting Models

1) SDXL Inpainting Model [3]: Initially, our pipeline was integrated with a baseline SDXL inpainting model. By taking the edited prompt and target mask as inputs, the model facilitates pixel-level image editing, ensuring precise modifications within the designated areas.

2) RFNet-Edit: The proposed RFNet-Edit framework, built upon the RFNet generative architecture, consists of two specialized stages: In-Depth Object Generation and Seamless Background Integration.

In the first stage, In-Depth Object Generation, the model utilizes a detailed textual description and a bounding box layout derived from the pipeline. Each target object is generated individually and constrained by its specified bounding box. This process employs localized prompts to ensure that object details and their predicted interactions with the environment remain semantically aligned with the edited prompt. Upon completion, the generated objects are positioned onto the original image according to the predefined layout.

The second stage, Seamless Background Integration, focuses on harmonizing the newly generated objects with the existing environment while preserving the integrity of both components. To define the transition zones, we employ the Segment Anything Model (SAM) to extract precise object masks. A composite mask is then created by forming a union of the background and identified object regions. During the denoising process, the latents corresponding to these masked areas are frozen, restricting the generative process to the bounding box boundary regions. By denoising only the remaining interfacial areas guided by the edited prompt, the model achieves a seamless transition between the generated objects and the original background.

Different from the SDXL inpainting model, RFNet-Edit operates in a mask-free manner. Users are only required to provide an edited prompt, while object localization and boundary handling are automatically inferred by the system.

III. Results

A. Experimental Setup

In this project, we choose Stable Diffusion XL as the text-to-image baseline model. We compare the performance of our method with Stable Diffusion XL inpainting model. The number of denoising steps is set as 50 with a fixed guidance scale of 7.5, and the synthetic images are in a resolution of 1024×1024.

B. Dataset

We evaluate our method on a subset of the EditBench [5] dataset, which is designed to assess text-guided image editing under complex and realistic conditions. EditBench contains natural-language editing instructions that involve multiple atomic actions, instance-level object manipulations, and explicit constraints on background preservation. From this dataset, we select 95 image–instruction pairs that require non-trivial object-level edits to modify multiple objects while maintaining overall scene consistency. To demonstrate the model's robustness across diverse visual domains, we curated a selection of original images encompassing various styles, including photorealistic and artistic/cartoon depictions. This diverse selection serves to validate the model's generalization capability and its effectiveness in maintaining stylistic consistency across varying image distributions. The testing subset provides a challenging benchmark for evaluating controllability and semantic alignment in image editing tasks.

C. Evaluation Metrics

We generate an image for each text prompt in our dataset selected from EditBench for evaluation. There are 95 images in total for evaluation. We selected the CLIP Score [8] for evaluation. The CLIP Score computes the cosine similarity between image and text embeddings produced by a pretrained CLIP model and is used to quantify the semantic alignment between the edited prompt and the generated image. Higher CLIP scores indicate better alignment between the generated output and the intended edit described by the prompt.

D. Quantitative Results

Our proposed framework, RFNet-Edit, significantly outperforms the baseline SDXL inpainting model in semantic alignment. Quantitatively, RFNet-Edit achieves a CLIP score of 27.41, compared to 25.69 for the SDXL inpainting model—a relative improvement of 6.67% using the same evaluation dataset.

While mask-based SDXL inpainting models allow for precise spatial localized editing, the rigid nature of mask constraints often limits the geometric flexibility of the target object, potentially leading to suboptimal or unnatural results. In contrast, our model utilizes bounding boxes, which grant the generative process greater topological flexibility. This approach allows the model to optimize the object's shape and posture within the designated area, resulting in superior visual quality and better adherence to the edited prompt.

E. Qualitative Results

Figure 1 compares RFNet-Edit with the SDXL inpainting baseline on representative examples from the EditBench dataset. The illustrated case corresponds to the instruction *"convert plant to flower, green to purple."* While the SDXL baseline only partially executes the instruction by changing the color of the pot and leaving the plant structure largely unchanged, RFNet-Edit correctly transforms the plant into flowers and applies the color modification as specified. Across similar examples, RFNet-Edit consistently demonstrates more accurate execution of multi-object edits and improved adherence to complex instructions, while preserving irrelevant background regions.

Figure 2 shows a qualitative comparison on an EditBench example with the instruction *"convert car to bicycle, kite to drone."* Although the SDXL inpainting baseline alters the appearance of the car, it fails to perform the required object replacement and does not modify the kite as instructed. In contrast, RFNet-Edit successfully converts the car into a bicycle and replaces the kite with a drone, demonstrating more accurate execution of multi-object edits and better adherence to the specified instruction.

Figure 3 and Figure 4 provide a qualitative comparison between the masks provided in the EditBench dataset and those generated by our proposed pipeline (RFNet-Edit). High-precision segmentation is a prerequisite for realistic editing; however, as shown in column (b), standard dataset masks often suffer from coarse boundaries and significant noise.

In Figure 3, the EditBench mask for the tennis racket and banana is fragmentary and includes artifacts that do not align with the object boundaries. In contrast, our approach (column c), which utilizes GroundingDINO for localization and SAM for refinement, successfully generates a sharp, dense mask that accurately captures the geometry of both the racket and the banana.

Similarly, in Figure 4, the target instruction requires isolating the pedestrian in the kimono. The EditBench mask is low-resolution and fails to separate the subject from the background clutter. Our method produces a crisp silhouette of the subject, ensuring that subsequent generative edits are strictly confined to the foreground subject without bleeding into the complex background. This improvement in mask fidelity directly contributes to the reduction of artifacts in the final inpainted results.

IV. Conclusion and discussion

A. Conclusion

In this paper, we presented RFNet-Edit, a structured, LLM-assisted framework designed to address the challenges of complex, multi-object image editing. Our approach bridges the gap between unstructured user intent and precise visual manipulation by implementing a modular pipeline. Specifically, we utilize Large Language Models (LLMs) for robust Instruction Parsing, enabling the system to decompose free-form text into structured, instance-level edit plans. To bridge the gap between text and vision, we incorporate Object Localization via open-vocabulary detection to define precise spatial coordinates , followed by Pixel-Level Segmentation using the Segment Anything Model (SAM) to ensure high-fidelity boundary definition. This integrated workflow allows for granular control over individual objects while mitigating the common pitfalls of rigid geometric constraints and background hallucinations seen in conventional models.

Central to our proposed framework is the RFNet-Edit generative architecture, which executes the editing process through two distinct phases. First, the In-Depth Object Generation stage synthesizes target objects individually within specified bounding boxes, using localized prompts to ensure semantic alignment. Unlike rigid mask-based approaches, this bounding-box constraint grants the model greater topological flexibility, allowing for necessary geometric transformations of the target object to better fit the user's intent. Subsequently, the Seamless Background Integration stage employs SAM to define precise transition zones. By freezing the latents of the preserved background and denoising only the interfacial areas, the model achieves a natural and cohesive blend between the newly generated content and the original scene.

Quantitative evaluations on the EditBench dataset demonstrate the efficacy of our method, with RFNet-Edit achieving a CLIP score of 27.41. This represents a 6.67% improvement over the baseline Stable Diffusion XL inpainting model, confirming enhanced semantic alignment and adherence to complex prompts. Furthermore, qualitative results across diverse visual domains, including photorealistic and artistic styles, validate the model's robustness and its ability to perform coordinated multi-object transformations while maintaining a seamless transition between edited elements and the original background. Overall, our findings underscore the importance of structured instruction interpretation and spatial awareness in achieving controllable, high-fidelity text-guided image editing.

B. Discussion

1) Other evaluation metrics: In addition to evaluating text-image alignment via CLIP Score, assessing the perceptual quality of the generated images is crucial. Initially, the Fréchet Inception Distance (FID) [9] was considered; however, due to the lack of paired ground-truth reference images in the EditBench dataset, standard distribution-based metrics were deemed inapplicable. Consequently, we introduced PickScore [10], a metric trained to predict human preference, to evaluate the generation quality.

In this assessment, our method achieved a PickScore of 20.8, which is comparable to, though slightly lower than, the SDXL inpainting baseline of 21.16. We attribute this marginal discrepancy to two primary factors. First, the limited sample size of the evaluation subset may introduce statistical variance impacting the mean score. Second, and more importantly, the SDXL baseline tends to adopt a conservative editing strategy, making minimal pixel-level modifications that largely preserve the high perceptual quality of the original source image. In contrast, our method performs more extensive structural and topological transformations (e.g., generating new object shapes via bounding boxes) to ensure semantic alignment. While this approach enhances instruction adherence, the complexity of integrating these generated structures with the background presents greater challenges, potentially resulting in minor

(a) Original image (b) SDXL inpainting result (c) RFNet-Edit result

Figure 1. Qualitative comparison between SDXL inpainting and RFNet-Edit on an EditBench example with the instruction *"convert plant to flower, green to purple,"* where RFNet-Edit correctly executes both object and color edits while SDXL only partially follows the instruction.

(a) Original image (b) SDXL inpainting result (c) RFNet-Edit result

Figure 2. Qualitative comparison between SDXL inpainting and RFNet-Edit on an EditBench example for the instruction *"convert car to bicycle, kite to drone."*

artifacts that slightly impact the preference score compared to the unmodified original regions preserved by SDXL.

2) Trade off between background preservation and target object flexibility: The SDXL inpainting model utilizes target masks to delineate editing regions, a method that excels at preserving the original background. However, this approach imposes strict geometric constraints on the newly generated object. When the target object requires a significant shape transformation—such as the "convert car to bicycle" example in Figure 2—the discrepancy between the original mask and the new object's topology often leads to generation failure. Conversely, our RFNet-Edit model employs bounding boxes, which decouple the generation process from the original object's shape, thereby ensuring higher success rates in complex conversions. This flexibility, however, introduces a challenge in seamless background integration, which remains the primary limitation of our method. Consequently, balancing background preservation with generative flexibility is critical. To mitigate integration artifacts, an optimal strategy involves minimizing the bounding box margin to be as tight as possible around the target object. This approach maximizes background retention while providing sufficient spatial freedom for the generation model to execute the intended transformation.

V. Future Work

While RFNet-Edit demonstrates improved performance on complex text-guided image editing tasks, several directions remain for further enhancing the robustness of the framework. One limitation stems from the inherent ambiguity of natural-language instructions, particularly when object references involve plural or collective expressions (e.g., cars, trees, or people). Such linguistic variability can lead to mismatches during open-vocabulary object localization.

A promising extension is the incorporation of plural-aware linguistic normalization during the object detection stage. By leveraging lightweight NLP toolkits such as SpaCy [7] to identify noun phrases and generate plural or group-based variants of target object queries, future versions of RFNet-Edit could improve recall for multi-instance objects without

(a) Original image (b) EditBench mask (c) GroundingDINO+SAM mask

Figure 3. Comparison of mask quality between EditBench ground truth and our GroundingDINO + SAM pipeline.

(a) Original image (b) EditBench mask (c) GroundingDINO+SAM mask

Figure 4. Comparison of mask quality between EditBench ground truth and our GroundingDINO + SAM pipeline.

requiring explicit user specification. Integrating this strategy with open-vocabulary detectors such as GroundingDINO may further reduce false negatives in crowded or complex scenes.

To further refine background integration, we propose optimizing bounding box margins alongside an advanced LLM-driven prompting strategy. Currently, the transition regions are denoised using the global edited prompt, which may lack the specificity required for seamless blending. A promising improvement involves leveraging the LLM to generate granular, context-aware descriptions specifically for the boundary zones. By guiding the diffusion model with these localized prompts rather than a generic global instruction, the system could achieve more coherent transitions between the generated object and the original scene.

For evaluation, our current study was restricted to a subset of 95 image-instruction pairs within a single category of the EditBench dataset due to computational resources constraints. With increased computational resources, we aim to scale this assessment across a broader spectrum of categories and more kinds of dataset to verify model generalization. Furthermore, we intend to extend the framework's capabilities to encompass a wider array of editing operations, such as object removal and scene-level background replacement, all unified under a single, streamlined natural-language interface.

We leave the systematic implementation of these algorithmic enhancements, specifically plural-aware target expansion and boundary-specific prompting, as well as comprehensive quantitative evaluation across diverse editing tasks, to future research.

VI. Statement of Individual Contribution

1) Terence Lin
   - **Model Development**: Designed and implemented the RFNet-Edit architecture, focusing on the two-stage generation and integration pipeline.
   - **Dataset Preparation**: Responsible for dataset collection and preprocessing from EditBench to ensure diverse stylistic representation.
   - **Architecture and experiment design**: Defined the overall project workflow and experiment setting.
2) Chelsea Chi

- **Natural Language Instruction Parsing with LLM**: Developed an LLM-based framework to translate natural language instructions into precise, machine-readable inputs.
- **Object Localization and Segmentation Mask Definition**: Constructed an experimental pipeline for text-to-region matching and pixel-level segmentation, reducing computational overhead while enabling multi-object editing.
- **Evaluation**: Generated baseline outputs and conducted evaluations for both the baseline and proposed methods.

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